

Chapitre 4. Structure from Motion

Prof. Slimane LARABI, USTHB

Chapitre 4. Structure from Motion

4.1 Overview

4.2 Structure from motion problem (SFM)

4.3 SFM Observation matrix

4.4 Rank of Observation Matrix

4.5 Tomasi-Kanade Factorization

Chapitre 4. Structure from Motion

4.1 Overview

The objective of structure from motion (sfm) is to be able to take a video of a scene and to recover the 3D structure of this scene.

From this acquired video, the proposed methods of the state of the art offer the possibility to extract the 3D structure of the scene so as the motion of the camera.

Note that the motion of the camera during the capture of the video is unknown.

We assume that the camera calibration is not needed because we will compute 3D points up to a scale factor.

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4.1 Overview

Example of video sequence of Roman ruins of Tipaza (Algeria)



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4.1 Overview

Example of video sequence of Roman ruins of Tipaza



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4.1 Overview

Example of video sequence of Roman ruins of Tipaza



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4.1 Overview

We can enumerate the possible applications of (sfm) as follow:

- Object Recognition
- Robotics
- Computer Graphics
- Image Retrieval
- Geo-Localization
- Archaeology
- Sports

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4.2 Problem of Structure from Motion

We suppose that we have a sequence video frames and features points such as corners, SIFT points are detected on each frame.

Also, we suppose that these features are tracked using known techniques: Template matching, Optical Flow (we know the correspondence between features of acquired images).



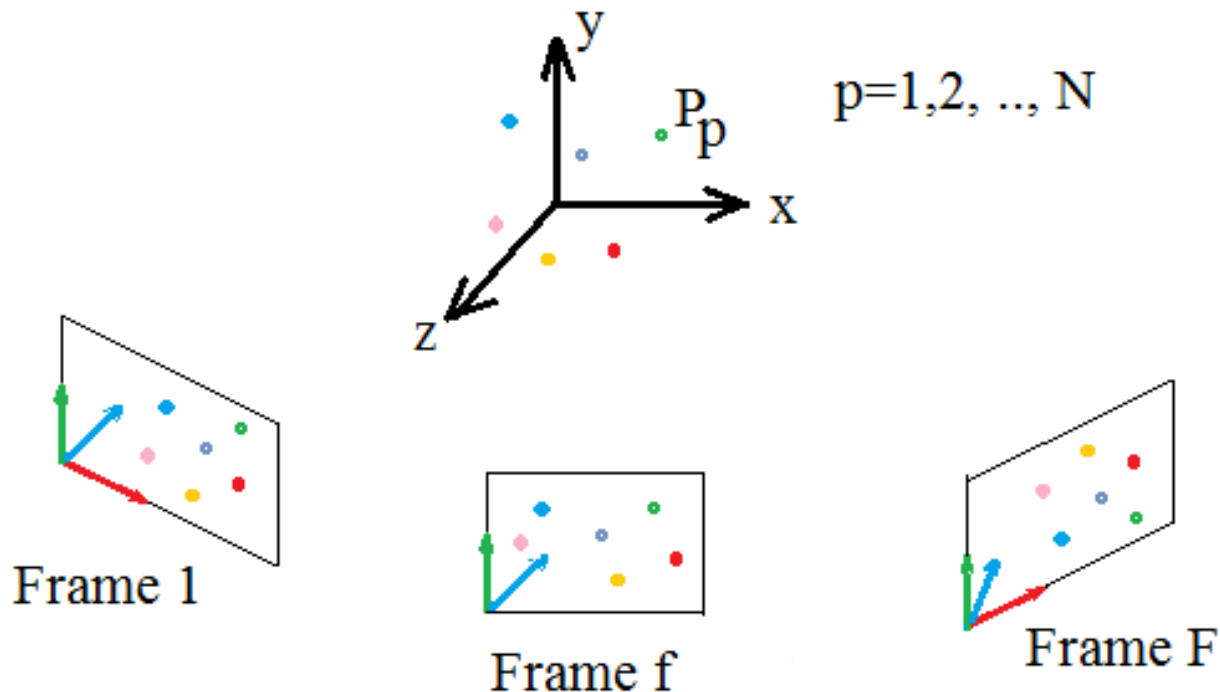
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4.2 Problem of Structure from Motion

Orthographic structure from motion

Let $(u_{f,p}, v_{f,p})$ a set of corresponding image points (2D) , where f is the frame number and p refers to the image point on that frame.

The main objective is to find the scene points (3D) assuming **orthographic camera**



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4.2 Problem of Structure from Motion

Orthographic structure from motion

Advantages of orthographic camera for structure from motion:

- simplifies the mathematical equations involved in the reconstruction process.
- the ray of projection of a 3D point is perpendicular onto the image plane, this simplifies the triangulation process.
- objects maintain their relative sizes regardless of their distance from the camera.
- It involves simpler mathematical operations, this implies faster computation times.
- It preserves the parallelism of lines and planar structures in the scene.

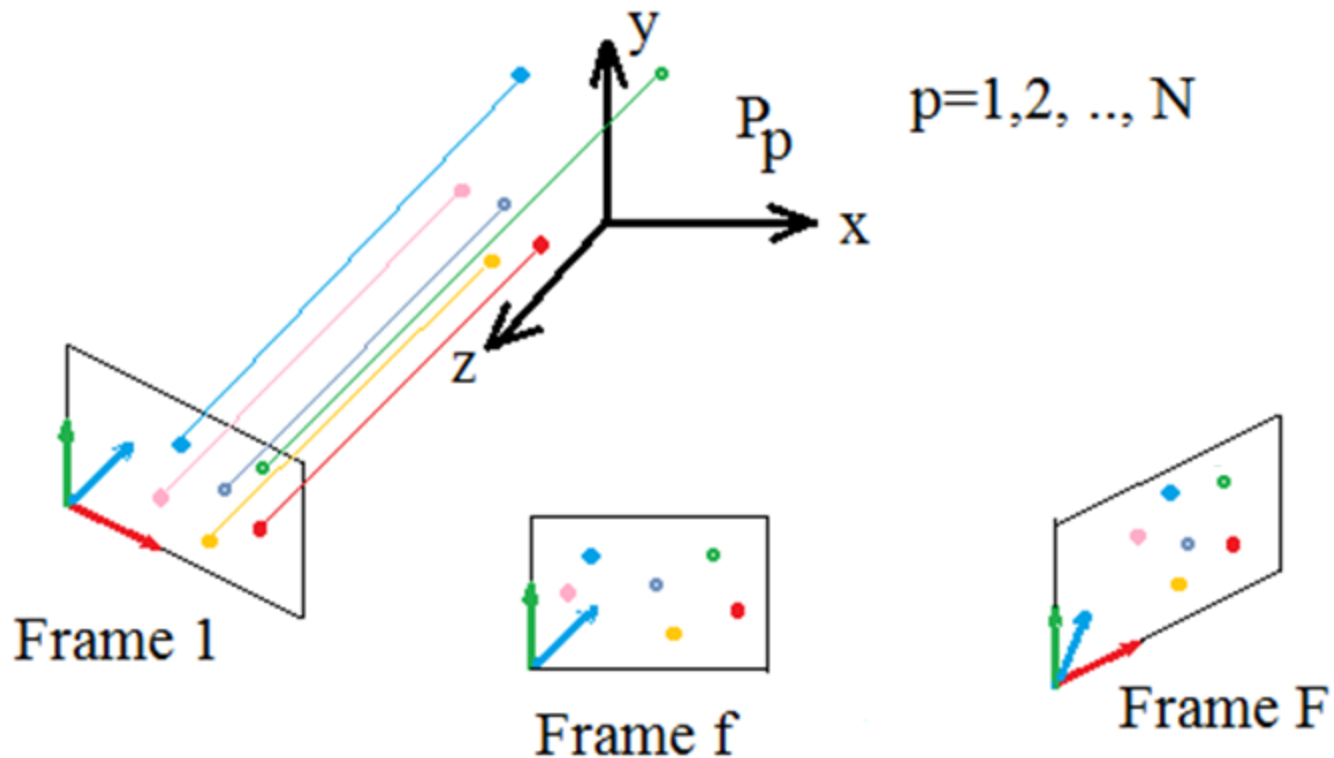
The limitations:

- It leads to inaccuracies in depth estimation,
- It is difficult to determine the absolute scale of the reconstructed scene without additional constraints or information.
- It leads to distortions and inaccuracies in the reconstructed 3D for scenes with irregular shapes or curved surfaces (non planar).

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4.2 Problem of Structure from Motion

Orthographic structure from motion

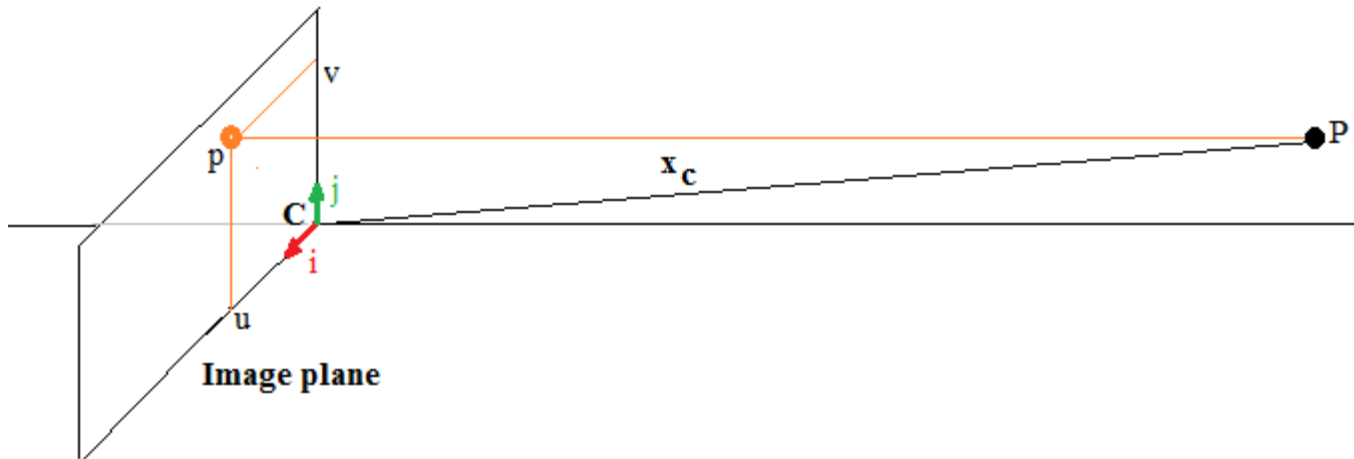


Orthographic Projection

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4.3 Observation Matrix

From 3D to 2D: Orthographic projection



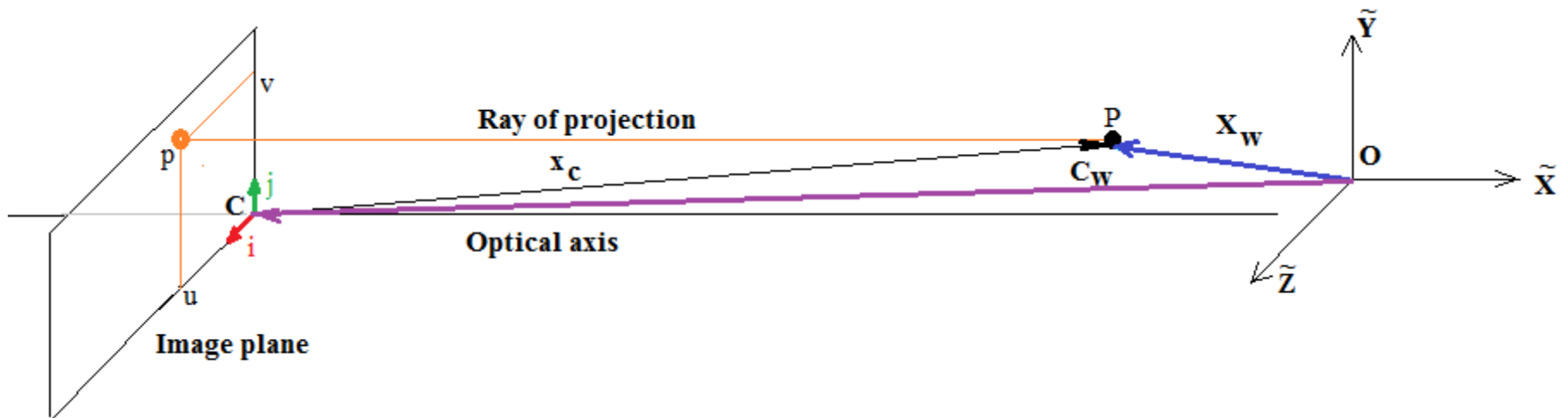
Perspective cameras exhibit orthographic projection where distance Of scene from camera is large compared to depth variation within scene (we can consider in this case planer surfaces).

$$u = i \cdot x_c (\text{scalar product}) = i^T x_c \qquad v = j \cdot x_c = j^T x_c$$

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4.3 Observation Matrix

From 3D to 2D: Orthographic projection



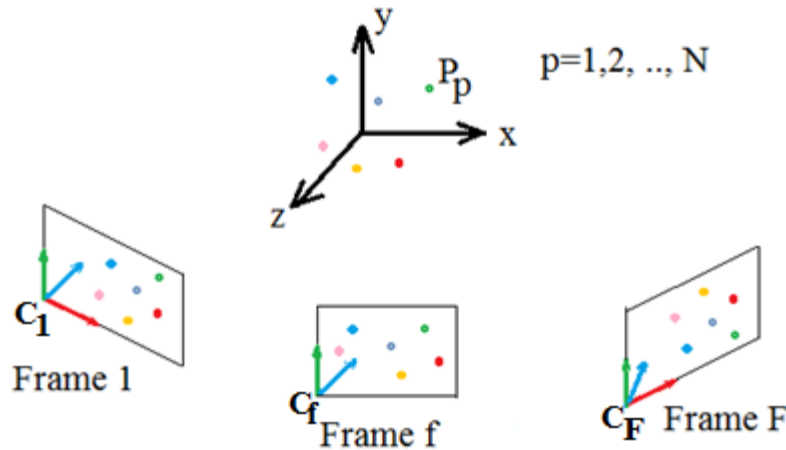
$$u = i \cdot x_c = i^T x_c = i^T (x_w - C_w) = i^T (P - C)$$

$$v = j \cdot x_c = j^T x_c = j^T (x_w - C_w) = j^T (P - C)$$

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4.3 Observation Matrix

Orthographic SFM



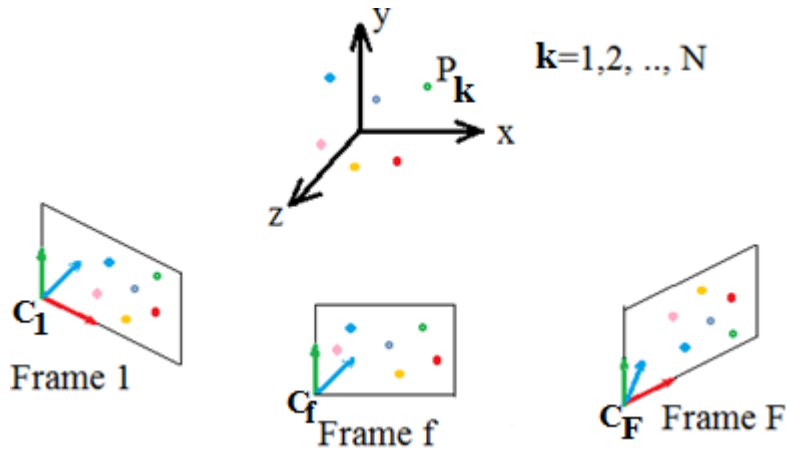
Let $(u_{f,p}, v_{f,p})$ a set of corresponding image points (2D) , where f is the frame number and $\mathbf{p}$ refers to the image point on that frame.

The main objective is to find the scene points (3D) assuming orthographic camera.

Camera positions (C_f) and orientations (i_f, j_f) are unknown.

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4.3 Observation Matrix



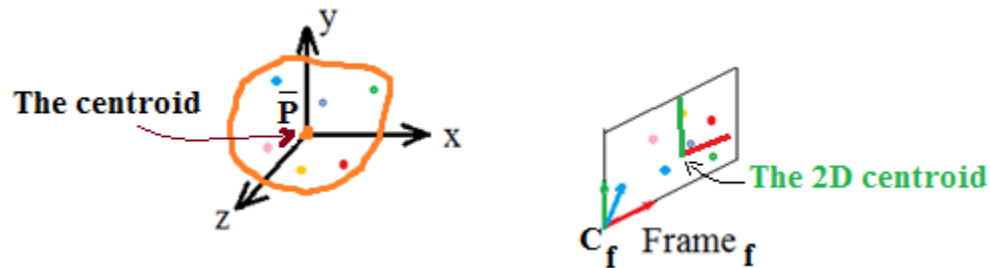
For each image point P_k in camera frame f we have:

$$u_{f,k} = i_f^T (P_k - C_f) \quad v_{f,k} = j_f^T (P_k - C_f)$$

We can remove C_f from equations to simplify SFM problem

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4.3 Observation Matrix

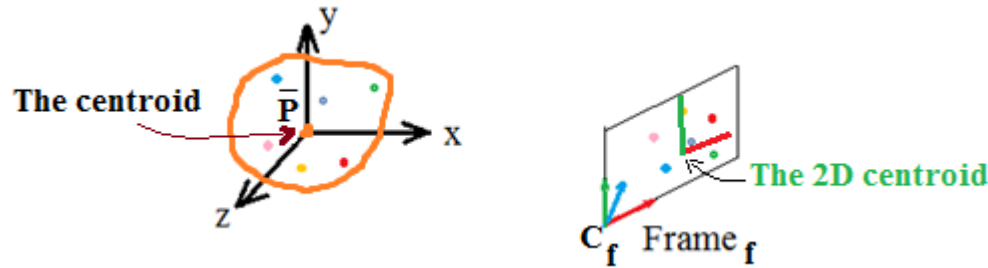


We assume the origin of the world at centroid of scene points.

$$\bar{P} = \frac{1}{N} \sum_{k=1}^N P_{f,k} = 0$$

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4.3 Observation Matrix



The centroid () of image points in frame f is given by:

$$\bar{u}_f = \frac{1}{N} \sum_{k=1}^N u_{f,k} = \frac{1}{N} \sum_{k=1}^N i_f^T (P_k - C_f) \quad \bar{v}_f = \frac{1}{N} \sum_{k=1}^N v_{f,k} = \frac{1}{N} \sum_{k=1}^N j_f^T (P_k - C_f)$$

$$\bar{u}_f = \frac{1}{N} i_f^T \sum_{k=1}^N P_k - \frac{1}{N} \sum_{k=1}^N i_f^T C_f = 0 - i_f^T C_f \quad \bar{v}_f = \frac{1}{N} j_f^T \sum_{k=1}^N P_k - \frac{1}{N} \sum_{k=1}^N j_f^T C_f = -j_f^T C_f$$

The centroid $(\bar{u}_f, \bar{v}_f)$ of feature points is not a function of the locations of scene points.

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4.3 Observation Matrix

We shift the origin to the centroid.
Image points w.r.t. $(\bar{u}_f, \bar{v}_f)$ are:

$$\bar{u}_{f,k} = u_{f,k} - \bar{u}_f$$

$$\bar{v}_{f,k} = v_{f,k} - \bar{v}_f$$

$$\bar{\mathbf{u}}_{f,k} = \mathbf{i}_f^T (\mathbf{P}_k - \mathbf{C}_f) + \mathbf{i}_f^T \mathbf{C}_f = \mathbf{i}_f^T \mathbf{P}_k \quad \bar{\mathbf{v}}_{f,k} = \mathbf{j}_f^T (\mathbf{P}_k - \mathbf{C}_f) + \mathbf{j}_f^T \mathbf{C}_f = \mathbf{j}_f^T \mathbf{P}_k$$

We can see that we have the image coordinates without the camera centre in the expression. Only the camera orientation $(\mathbf{i}_f^T, \mathbf{j}_f^T)$ and scene point are present ($\mathbf{C}_f$ removed).

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4.3 Observation Matrix

Point 1 Point 2 ...Point N

$$\mathbf{W} = \mathbf{M} \times \mathbf{S}$$

$$\begin{array}{l} \text{Image 1} \\ \dots \\ \text{Image F} \\ \text{Image 1} \\ \dots \\ \text{Image F} \end{array} \begin{bmatrix} \bar{u}_{1,1} & \bar{u}_{1,2} & \dots & \bar{u}_{1,N} \\ \vdots & & \ddots & \vdots \\ \bar{u}_{F,1} & \bar{u}_{F,2} & \dots & \bar{u}_{F,N} \\ \bar{v}_{1,1} & \bar{v}_{1,2} & \dots & \bar{v}_{1,N} \\ \vdots & & \ddots & \vdots \\ \bar{v}_{F,1} & \bar{v}_{F,2} & \dots & \bar{v}_{F,N} \end{bmatrix} = \begin{bmatrix} i_1^T \\ i_2^T \\ \vdots \\ i_F^T \\ j_1^T \\ j_2^T \\ \vdots \\ j_F^T \end{bmatrix} \begin{bmatrix} P_1 & P_2 & \dots & P_N \end{bmatrix}$$

$\mathbf{S}(3 \times N)$: Unknown 3D structure

$\mathbf{M}(2F \times 3)$: Unknown Camera motion

$\mathbf{W}(2F \times N)$: Known Centroid-subtracted Feature Points

How to find $\mathbf{M}$ and $\mathbf{S}$ from $\mathbf{W}$?

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4.4 Rank of Observation Matrix

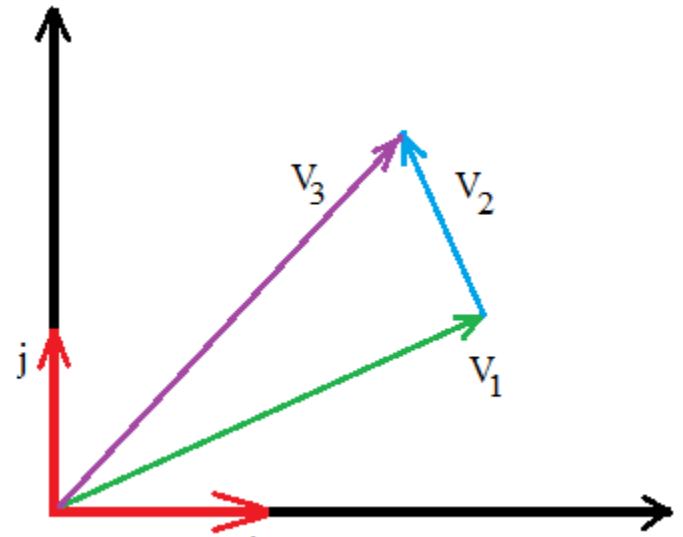
A set of vectors $(V_1, V_2, \dots, V_n)$ are linearly independent if no vector V_i can be represented as a weighted linear sum of the others.

Examples:

(i, j) are linearly independent

(i, j, V_1) , (i, j, V_3) , are linearly dependent

(V_1, V_2, V_3) are linearly dependent



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4.4 Rank of Observation Matrix

We will note:

Column Rank: the number of linearly independent columns of the matrix.

Row Rank: the number of linearly independent rows of the matrix.

$$\begin{matrix} m \\ \left[\begin{array}{c} A \end{array} \right] \\ n \end{matrix} = [c_1 \ c_2 \ \dots \ c_n] = \begin{bmatrix} r_1^T \\ r_2^T \\ \vdots \\ \vdots \\ r_m^T \end{bmatrix}$$

Column Rank(A) ≤ n, Row Rank(A) ≤ m

Column Rank(A) = Row Rank(A) = Rank(A)

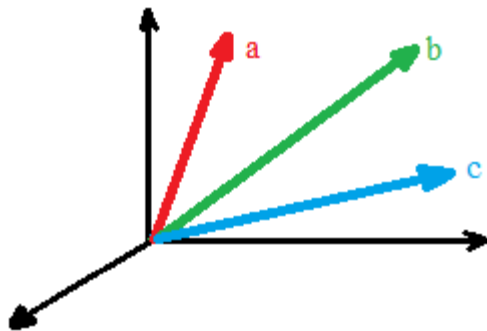
Rank(A) ≤ min(m, n)

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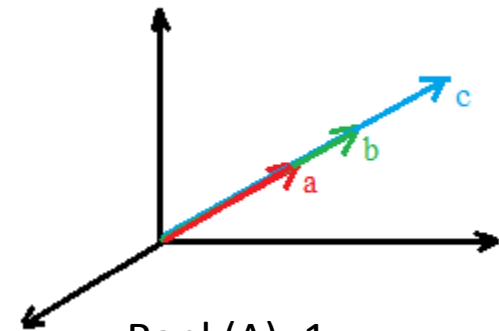
4.4 Rank of Observation Matrix

Rank is the dimensionality of the space spanned by column or row vectors of the matrix

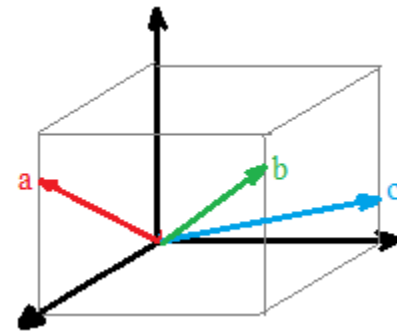
$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = [a \quad b \quad c]$$



Rank(A)=2



Rank(A)=1



Rank(A)=3

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4.4 Rank of Observation Matrix

$$\text{Rank}(A^T) = \text{Rank}(A)$$

$$\begin{aligned}\text{Rank}(A_{m \times n} B_{n \times p}) &= \min(\text{Rank}(A_{m \times n}), \text{Rank}(B_{n \times p})) \\ &\leq \min(m, n, p)\end{aligned}$$

$$\text{Rank}(AA^T) = \text{Rank}(A^T A) = \text{Rank}(A) = \text{Rank}(A^T)$$

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4.5 Tomasi-Kanade Factorization

Tomasi, Carlo. & Kanade, Takeo. 1992 Shape and motion from image streams under orthography: a factorization method. Int. J. Comp. Vision 9, 137-54.



Tomasi, Carlo



Kanade, Takeo

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4.5 Tomasi-Kanade Factorization

Tomasi, Carlo. & Kanade, Takeo. 1992 Shape and motion from image streams under orthography: a factorization method. Int. J. Comp. Vision 9, 137-54.

$$\begin{matrix} W & = & M & \times & S \\ 2F \times N & & 2F \times 3 & & 3 \times N \end{matrix}$$

We know:

$$\begin{aligned} \text{Rank}(MS) &\leq \text{Rank}(M), & \text{Rank}(MS) &\leq \text{Rank}(S) \\ \text{Rank}(MS) &\leq \min(3, 2F), & \text{Rank}(MS) &\leq \min(3, N) \\ \text{Rank}(W) &= \text{Rank}(MS) \leq \min(3, N, 2F) \end{aligned}$$

Rank Theorem: $\text{Rank}(W) \leq 3$

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4.5 Tomasi-Kanade Factorization

Singular Value Decomposition (SVD)

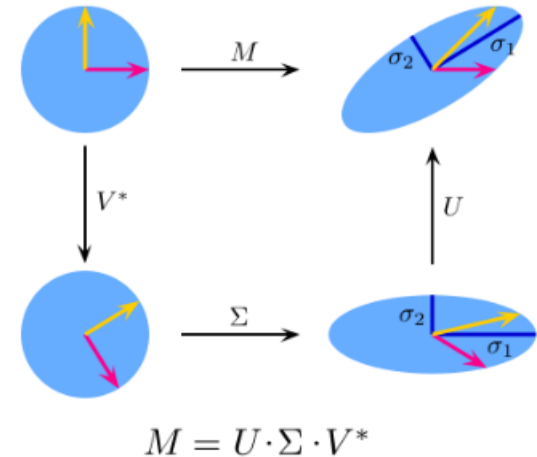
For any matrix A there exists a factorization:

$$A_{M \times N} = U_{M \times M} \sum_{M \times N} V_{N \times N}^T$$

Where U and V^T are orthonormal and Σ is diagonal

$$\sum_{M \times N} = \begin{bmatrix} \sigma_1 & 0 & 0 & 0 & 0 & - & 0 \\ 0 & \sigma_2 & 0 & 0 & 0 & - & 0 \\ 0 & 0 & \sigma_3 & 0 & 0 & - & 0 \\ - & - & - & - & - & - & - \\ 0 & 0 & 0 & 0 & 0 & - & \sigma_N \\ 0 & 0 & 0 & 0 & 0 & - & 0 \end{bmatrix}$$

$\sigma_1, \sigma_2, \dots, \sigma_N$: Singular values



If **Rank(A)=r** , then A has **r** non-zero singular values.

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4.5 Tomasi-Kanade Factorization

Singular Value Decomposition (SVD)

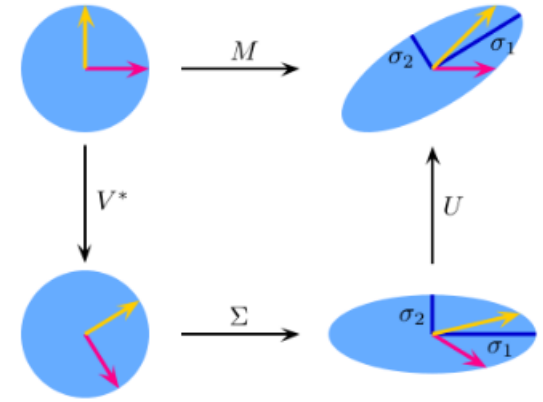
$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 \end{pmatrix}$$

$$U = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 4 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$V^* = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1/\sqrt{5} & 0 & 0 & 0 & 2/\sqrt{5} \\ 0 & 0 & 0 & 1 & 0 \\ -2/\sqrt{5} & 0 & 0 & 0 & 1/\sqrt{5} \end{pmatrix}$$

$$UU^* = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$VV^* = \begin{pmatrix} 0 & 0 & 1/\sqrt{5} & 0 & -2/\sqrt{5} \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2/\sqrt{5} & 0 & 1/\sqrt{5} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1/\sqrt{5} & 0 & 0 & 0 & 2/\sqrt{5} \\ 0 & 0 & 0 & 1 & 0 \\ -2/\sqrt{5} & 0 & 0 & 0 & 1/\sqrt{5} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$



$$M = U \cdot \Sigma \cdot V^*$$

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4.5 Tomasi-Kanade Factorization

Singular Value Decomposition (SVD)

As $\text{Rank}(W) \leq 3$, then:

$$W = U \sum V^T$$

And $\sigma_i = 0$ for $i = 4..N$

$$\sum = \begin{bmatrix} \sigma_1 & 0 & 0 & 0 & 0 & - & 0 \\ 0 & \sigma_2 & 0 & 0 & 0 & - & 0 \\ 0 & 0 & \sigma_3 & 0 & 0 & - & 0 \\ 0 & 0 & 0 & 0 & 0 & - & 0 \\ 0 & 0 & 0 & 0 & 0 & - & 0 \end{bmatrix}$$

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4.5 Tomasi-Kanade Factorization

$$W = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & 0 & 0 & 0 & - & 0 \\ 0 & \sigma_2 & 0 & 0 & 0 & - & 0 \\ 0 & 0 & \sigma_3 & 0 & 0 & - & 0 \\ 0 & 0 & 0 & 0 & 0 & - & 0 \\ 0 & 0 & 0 & 0 & 0 & - & 0 \end{bmatrix} \begin{bmatrix} V_1^T \\ V_2^T \end{bmatrix}$$

$\begin{matrix} 3 & 2F-3 & 2FxN & NxN \end{matrix}$

Since $\text{Rank}(W) \leq 3$, $\text{Rank}(\Sigma) \leq 3$, then submatrices U_2 and V_2^T do not contribute to W

$$W = U_1 \sum_1 V_1^T$$

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4.5 Tomasi-Kanade Factorization

Factorization (Finding M, S)

$$W = U_1 \left(\sum_1 \right)^{\frac{1}{2}} \left(\sum_1 \right)^{1/2} V_1^T$$

For any 3x3 non-singular matrix Q, the following equation is also valid.

$$W = U_1 \left(\sum_1 \right)^{\frac{1}{2}} Q Q^{-1} \left(\sum_1 \right)^{1/2} V_1^T$$

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4.5 Tomasi-Kanade Factorization

Orthonormality of M:

The matrix M:

$$M = \begin{bmatrix} i_1^T \\ i_2^T \\ i_F^T \\ j_1^T \\ j_2^T \\ j_F^T \end{bmatrix} = U_1 \underbrace{(\Sigma_1)^{1/2}}_{\text{Computed}} Q = \begin{bmatrix} \hat{i}_1^T \\ \hat{i}_2^T \\ \hat{i}_F^T \\ \hat{j}_1^T \\ \hat{j}_2^T \\ \hat{j}_F^T \end{bmatrix} Q = \begin{bmatrix} \hat{i}_1^T Q \\ \hat{i}_2^T Q \\ \hat{i}_F^T Q \\ \hat{j}_1^T Q \\ \hat{j}_2^T Q \\ \hat{j}_F^T Q \end{bmatrix}$$

Computed

Computed

Orthonormality constraints:

$$\hat{i}_f^T \hat{i}_f^T = 1, \hat{j}_f^T \hat{j}_f^T = 1, \hat{i}_f^T \hat{j}_f^T = 0,$$

Then, for one frame we get three equations:

$$\hat{i}_f^T Q Q^T \hat{i}_f^T = 1, \hat{j}_f^T Q Q^T \hat{j}_f^T = 1, \hat{i}_f^T Q Q^T \hat{j}_f^T = 0,$$

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4.5 Tomasi-Kanade Factorization

We computed $(\hat{i}_f^T, \hat{j}_f^T)$ for $f=1, \dots, F$
 $\hat{i}_f^T Q Q^T \hat{i}_f^T = 1, \hat{j}_f^T Q Q^T \hat{j}_f^T = 1, \hat{i}_f^T Q Q^T \hat{j}_f^T = 0$, Q is unknown.

Q is 3×3 matrix, 9 variables, $3F$ quadratic equations.

Q can be solved with 3 or more images ($F \geq 3$) using Newton's method

Final solution:

$$M = U_1 (\mathbf{\Sigma}_1)^{1/2} Q \quad \text{and} \quad S = Q^{-1} \mathbf{\Sigma}_1^{1/2} V_1^T$$

Camera motion Scene structure

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4.5 Tomasi-Kanade Factorization

Summary: Orthographic SFM

- 1- Detect and track feature points
- 2- Create the centroid subtracted matrix W of corresponding feature points
- 3- Compute SVD of W and enforce rank constraint.

$$W = U \Sigma V^T = U_1 \Sigma_1 V_1^T$$

- 4- Set $M = U_1 (\Sigma_1)^{1/2} Q$ and $S = Q^{-1} \Sigma_1^{1/2} V_1^T$

- 5- Find Q by enforcing the orthornormality constraint

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4.5 Tomasi-Kanade Factorization

Example of Tomasi 1992, Duke University, USA



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4.5 Tomasi-Kanade Factorization

Implementation of Marc Pollefeys 2002 (ETH Zurich)



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4.5 Tomasi-Kanade Factorization

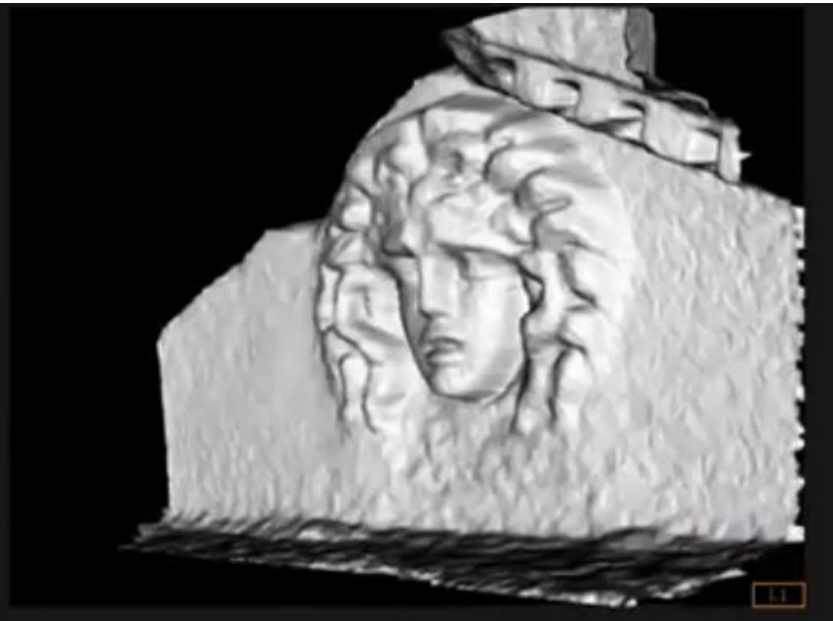
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4.5 Tomasi-Kanade Factorization

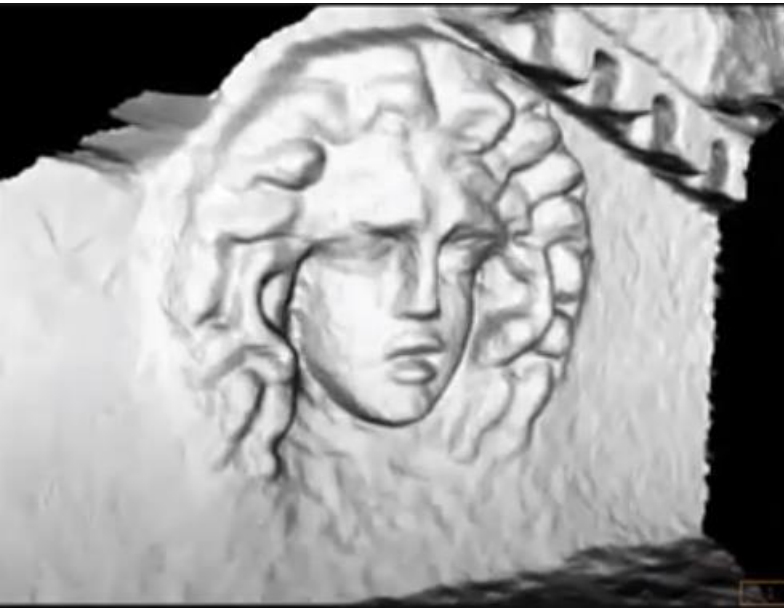
Implementation of Marc Pollefeys 2002 (ETH Zurich)



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4.5 Tomasi-Kanade Factorization

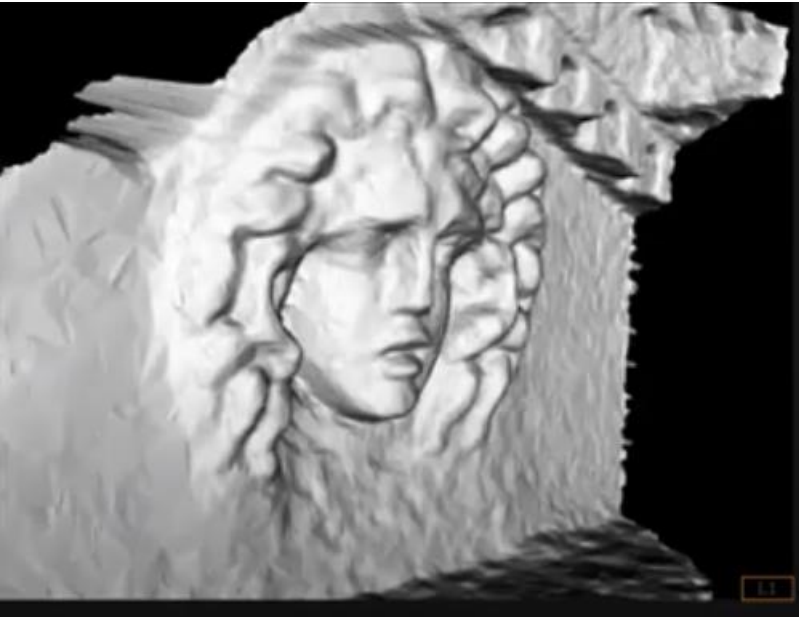
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